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Subject : Python for Visual AI

Assignment 7:

An Agri-analytics company wants to collect daily vegetable prices from government portals.https://agmarknet.gov.in/

Step 1: Importing the necessary python libraries

In [1]:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
```

Step 2: Loading the dataset into pandas dataframe

In [2]:

```
df=pd.read_csv("Marketwise_Price_Arrival_01-04-2026_11-22-48_PM.csv",skiprows=2)
df
```

Out[2]:

	Commodity Group	Commodity	MSP (Rs./Quintal) 2026-27	Price on 30 Mar, 2026	Price on 29 Mar, 2026	Price on 28 Mar, 2026	Arrival on 30 Mar, 2026	A
0	Cereals	Bajra(Pearl Millet/Cumbu)	2775.00	2362.87	1855.71	2550.95	413.87	4
1	Cereals	Barley(Jau)	2150.00	1987.90	1931.55	1962.55	1777.83	67
2	Cereals	Jowar(Sorghum)	3699.00	4637.66	2374.25	4132.76	149.96	
3	Cereals	Maize	2400.00	1698.31	1688.53	1739.70	18271.32	75
4	Cereals	Paddy(Common)	2369.00	2689.67	2470.59	2969.77	9803.90	94
5	Cereals	Ragi(Finger Millet)	4886.00	4328.97	3126.82	3533.31	54.20	5
6	Cereals	Wheat	2585.00	2389.95	2297.05	2479.65	49768.73	144
7	Fibre Crops	Cotton	7710.00	7783.73	7435.72	8095.73	3520.05	33
8	Oil Seeds	Copra	12100.00	22324.47	20749.88	17375.02	69.93	

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	Commodity Group	Commodity	MSP (Rs./Quintal) 2026-27	Price on 30 Mar, 2026	Price on 29 Mar, 2026	Price on 28 Mar, 2026	Arrival on 30 Mar, 2026	A
9	Oil Seeds	Groundnut	7263.00	7084.98	6694.92	7278.72	5635.54	
10	Oil Seeds	Mustard	6200.00	6570.21	6411.53	6461.71	10693.51	81
11	Oil Seeds	Niger Seed(Ramtil)	9537.00	11250.00	-	10500.00	0.30	
12	Oil Seeds	Safflower	6540.00	-	-	4643.43	-	
13	Oil Seeds	Sesamum(Sesame,Gingelly,Til)	9846.00	7823.08	7496.55	11986.87	5.20	
14	Oil Seeds	Soyabean	5328.00	5147.72	5182.65	5380.13	3640.82	7
15	Oil Seeds	Sunflower	7721.00	6624.26	6483.00	6593.24	178.40	1
16	Others	Sugarcane	285.00	331.71	300.00	300.00	7025.10	
17	Pulses	Arhar(Tur/Red Gram)(Whole)	8000.00	7632.36	7390.55	7600.05	1355.21	7
18	Pulses	Bengal Gram(Gram)(Whole)	5875.00	5386.13	5174.37	5393.93	4371.52	13
19	Pulses	Black Gram(Urd Beans)(Whole)	7800.00	8528.49	7511.45	8110.81	220.00	
20	Pulses	Green Gram(Moong)(Whole)	8768.00	8906.90	6558.14	8545.69	515.42	
21	Pulses	Lentil(Masur)(Whole)	7000.00	7792.09	6762.74	8087.63	550.88	
22	Vegetables	Onion	-	1271.47	1158.03	1212.85	7575.46	29
23	Vegetables	Potato	-	738.41	951.64	834.45	12753.84	50
24	Vegetables	Tomato	-	1284.93	1387.78	1286.17	3738.68	16

Step 3 : Data Preprocessing

In [3]:

```
df.head()
```

Out[3]:

	Commodity Group	Commodity	MSP (Rs./Quintal) 2026-27	Price on 30 Mar, 2026	Price on 29 Mar, 2026	Price on 28 Mar, 2026	Arrival on 30 Mar, 2026	Arrival on 29 Mar, 2026	Arrival on 28 Mar, 2026
0	Cereals	Bajra(Pearl Millet/Cumbu)	2775.00	2362.87	1855.71	2550.95	413.87	405.00	531.85
1	Cereals	Barley(Jau)	2150.00	1987.90	1931.55	1962.55	1777.83	6739.82	1473.69
2	Cereals	Jowar(Sorghum)	3699.00	4637.66	2374.25	4132.76	149.96	48.00	186.90
3	Cereals	Maize	2400.00	1698.31	1688.53	1739.70	18271.32	7520.58	12325.91
4	Cereals	Paddy(Common)	2369.00	2689.67	2470.59	2969.77	9803.90	9443.77	14498.04

In [4]:

```
df.info()
```

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```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25 entries, 0 to 24
Data columns (total 9 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   Commodity Group             25 non-null     object
 1   Commodity                   25 non-null     object
 2   MSP (Rs./Quintal) 2026-27   25 non-null     object
 3   Price on 30 Mar, 2026       25 non-null     object
 4   Price on 29 Mar, 2026       25 non-null     object
 5   Price on 28 Mar, 2026       25 non-null     float64
 6   Arrival on 30 Mar, 2026     25 non-null     object
 7   Arrival on 29 Mar, 2026     25 non-null     object
 8   Arrival on 28 Mar, 2026     25 non-null     float64
dtypes: float64(2), object(7)
memory usage: 1.9+ KB
```

In [5]:

```
df.isnull().sum()
```

Out[5]:

```
Commodity Group              0
Commodity                    0
MSP (Rs./Quintal) 2026-27    0
Price on 30 Mar, 2026        0
Price on 29 Mar, 2026        0
Price on 28 Mar, 2026        0
Arrival on 30 Mar, 2026      0
Arrival on 29 Mar, 2026      0
Arrival on 28 Mar, 2026      0
dtype: int64
```

Step 4 : Renaming Columns

In [6]:

```
df.rename(columns={
    'msp_(rs./quintal)_2026-27': 'msp',
    'price_on_30_mar,_2026': 'price_30',
    'price_on_29_mar,_2026': 'price_29',
    'price_on_28_mar,_2026': 'price_28',
    'arrival_on_30_mar,_2026': 'arrival_30',
    'arrival_on_29_mar,_2026': 'arrival_29',
    'arrival_on_28_mar,_2026': 'arrival_28'
}, inplace=True)
```

Step 5 : Convert categorical columns into numeric

In [7]:

```
cols = [
    'MSP (Rs./Quintal) 2026-27',
    'Price on 30 Mar, 2026',
    'Price on 29 Mar, 2026',
    'Price on 28 Mar, 2026',
    'Arrival on 30 Mar, 2026',
```

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```
    'Arrival on 29 Mar, 2026',
    'Arrival on 28 Mar, 2026'
]

for col in cols:
    df[col] = pd.to_numeric(df[col], errors='coerce')
```

Step 6 : MSP Comparison

The market price is compared with MSP to determine farmer profit.

$$Above\ MSP = \begin{cases} 1 & \text{if Price} > \text{MSP} \\ 0 & \text{otherwise} \end{cases}$$

In [8]:

```
df['above_msp_30'] = df['Price on 30 Mar, 2026'] > df['MSP (Rs./Quintal) 2026-27']
df['above_msp_29'] = df['Price on 29 Mar, 2026'] > df['MSP (Rs./Quintal) 2026-27']
df['above_msp_28'] = df['Price on 28 Mar, 2026'] > df['MSP (Rs./Quintal) 2026-27']
```

Step 7 : Average Price Calculation

Average price is calculated to understand overall trends.

$$AveragePrice = \frac{\sum_{i=1}^{n} Price_i}{n}$$

In [9]:

```
print("Avg Price 30 March:\n", df['Price on 30 Mar, 2026'].mean())
print("Avg Price 29 March:\n", df['Price on 29 Mar, 2026'].mean())
print("Avg Price 28 March:\n", df['Price on 28 Mar, 2026'].mean())

print(df.groupby('Commodity')[['Price on 30 Mar, 2026','Price on 29 Mar, 2026','Price on
```

```
Avg Price 30 March:
 5690.6779166666665
Avg Price 29 March:
 4930.15
Avg Price 28 March:
 5562.204800000001
                            Price on 30 Mar, 2026  Price on 29 Mar, 2026  \
Commodity
Arhar(Tur/Red Gram)(Whole)                 7632.36                7390.55
Bajra(Pearl Millet/Cumbu)                  2362.87                1855.71
Barley(Jau)                                1987.90                1931.55
Bengal Gram(Gram)(Whole)                   5386.13                5174.37
Black Gram(Urd Beans)(Whole)               8528.49                7511.45
Copra                                     22324.47               20749.88
Cotton                                     7783.73                7435.72
Green Gram(Moong)(Whole)                   8906.90                6558.14
Groundnut                                  7084.98                6694.92
Jowar(Sorghum)                             4637.66                2374.25
```

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Lentil(Masur)(Whole)	7792.09	6762.74
Maize	1698.31	1688.53
Mustard	6570.21	6411.53
Niger Seed(Ramtil)	11250.00	NaN
Onion	1271.47	1158.03
Paddy(Common)	2689.67	2470.59
Potato	738.41	951.64
Ragi(Finger Millet)	4328.97	3126.82
Safflower	NaN	NaN
Sesamum(Sesame,Gingelly,Til)	7823.08	7496.55
Soyabean	5147.72	5182.65
Sugarcane	331.71	300.00
Sunflower	6624.26	6483.00
Tomato	1284.93	1387.78
Wheat	2389.95	2297.05

Commodity	Price on 28 Mar, 2026
Arhar(Tur/Red Gram)(Whole)	7600.05
Bajra(Pearl Millet/Cumbu)	2550.95
Barley(Jau)	1962.55
Bengal Gram(Gram)(Whole)	5393.93
Black Gram(Urd Beans)(Whole)	8110.81
Copra	17375.02
Cotton	8095.73
Green Gram(Moong)(Whole)	8545.69
Groundnut	7278.72
Jowar(Sorghum)	4132.76
Lentil(Masur)(Whole)	8087.63
Maize	1739.70
Mustard	6461.71
Niger Seed(Ramtil)	10500.00
Onion	1212.85
Paddy(Common)	2969.77
Potato	834.45
Ragi(Finger Millet)	3533.31
Safflower	4643.43
Sesamum(Sesame,Gingelly,Til)	11986.87
Soyabean	5380.13
Sugarcane	300.00
Sunflower	6593.24
Tomato	1286.17
Wheat	2479.65

Step 8 : Group-wise Analysis

Prices are grouped by commodity.

$$\bar{P}_{commodity} = \frac{\sum Price}{Count}$$

In [10]:

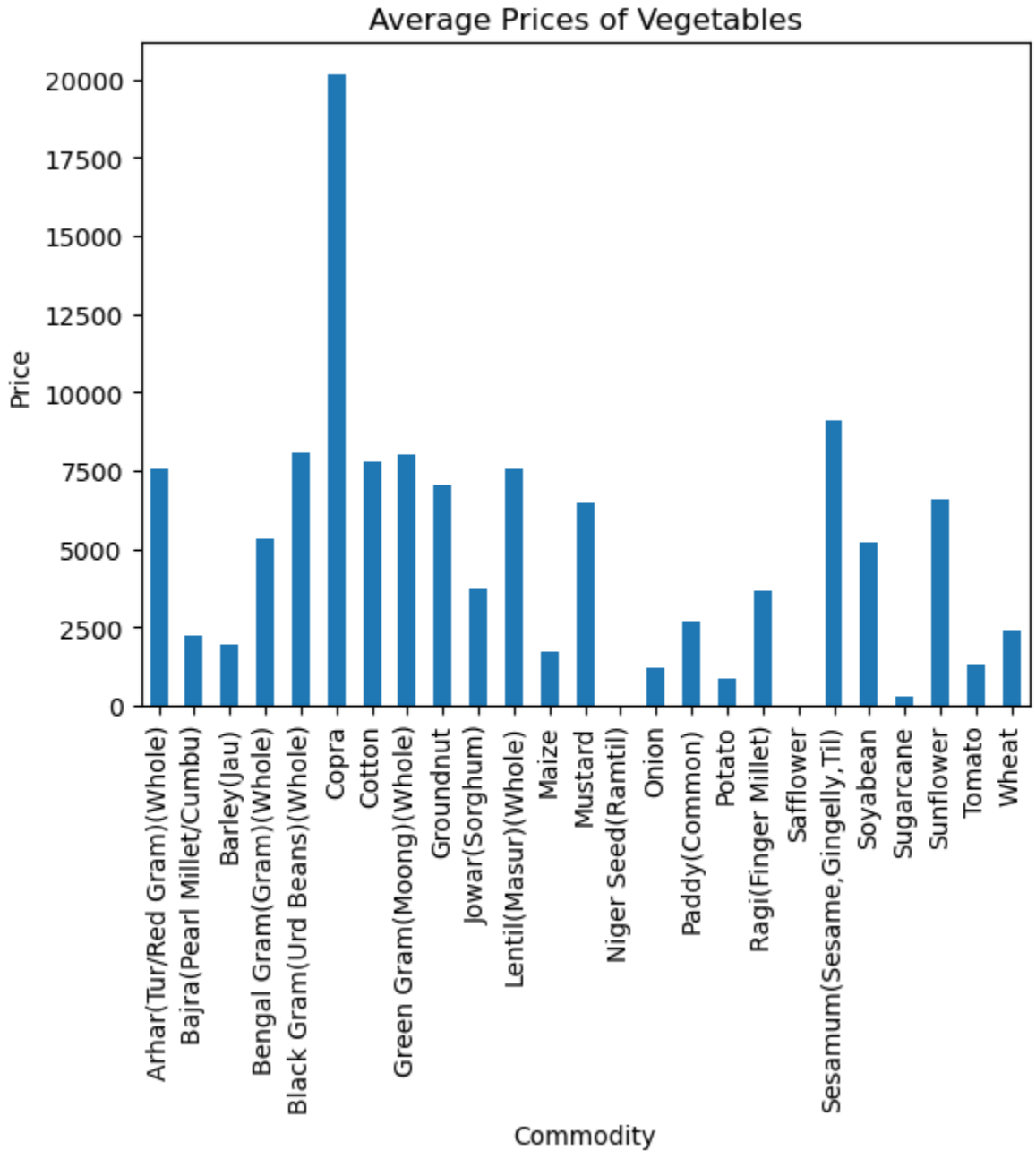
```
df_final = df.copy()
df_final['Avg Price'] = (
    df_final['Price on 30 Mar, 2026'].astype(float) +
```

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```
    df_final['Price on 29 Mar, 2026'].astype(float) +
    df_final['Price on 28 Mar, 2026']
) / 3

df_final.groupby('Commodity')['Avg Price'].mean().plot(kind='bar')

plt.title("Average Prices of Vegetables")
plt.xlabel("Commodity")
plt.ylabel("Price")
plt.show()
```



Step 9 : Price Trend Analysis

Price trend shows variation over time.

$$Trend = f(\text{Time})$$

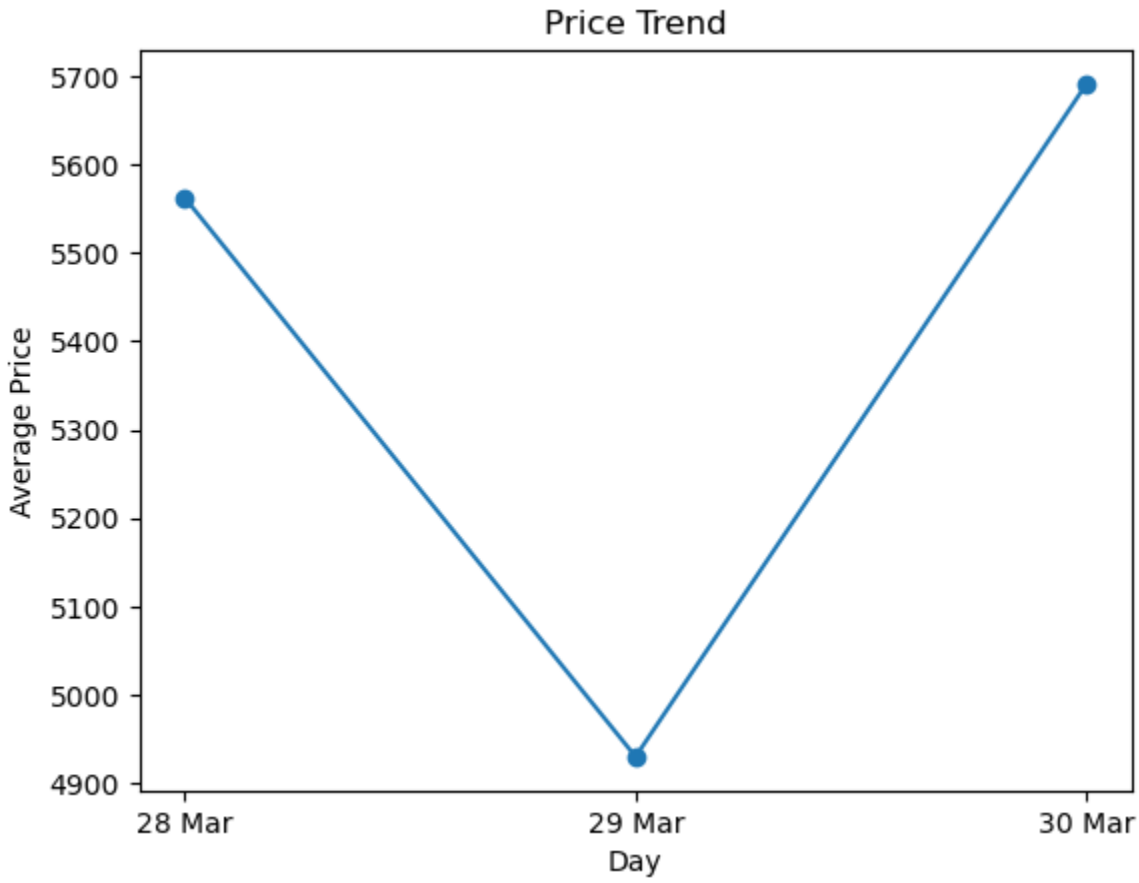
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In [11]:

```
avg_prices = [
    df['Price on 28 Mar, 2026'].mean(),
    df['Price on 29 Mar, 2026'].mean(),
    df['Price on 30 Mar, 2026'].mean()
]

days = ['28 Mar', '29 Mar', '30 Mar']

plt.plot(days, avg_prices, marker='o')
plt.title("Price Trend")
plt.xlabel("Day")
plt.ylabel("Average Price")
plt.show()
```



Step 10 : Building a Model

Linear Regression Model

Linear regression is used for price prediction.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$$

Model Explanation

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The linear regression model is given by:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$$

where,

y =Predicted Price

x_1 = MSP (Minimum Support Price)

x_2 =Price on previous day

x_3 =Arrival of commodity

$$\beta_0, \beta_1, \beta_2, \beta_3 \ = \ \text{Model coefficients}$$

Prediction

The model predicts future prices.

$$\hat{y} = f(X)$$

In [12]:

```
import pandas as pd
from sklearn.linear_model import LinearRegression

# Convert columns to numeric
cols = [
    'MSP (Rs./Quintal) 2026-27',
    'Price on 28 Mar, 2026',
    'Price on 29 Mar, 2026',
    'Arrival on 30 Mar, 2026',
    'Price on 30 Mar, 2026'
]

for col in cols:
    df[col] = pd.to_numeric(df[col], errors='coerce')

# Define X and y
X = df[['MSP (Rs./Quintal) 2026-27',
        'Price on 28 Mar, 2026',
        'Price on 29 Mar, 2026',
        'Arrival on 30 Mar, 2026']]

y = df['Price on 30 Mar, 2026']

# Remove missing values
valid_indices = X.dropna().index
X = X.loc[valid_indices]
y = y.loc[valid_indices]

# Train model
model = LinearRegression()
model.fit(X, y)

#  Prediction with proper feature names
new_data = pd.DataFrame([[2500, 2000, 2100, 500]],
```

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```
                          columns=X.columns)

prediction = model.predict(new_data)

print("Predicted Price:", prediction[0])
```

```
Predicted Price: 2545.0778172372106
```

Step 11 : Saving File

In [13]:

```
df.to_csv("final_dataset.csv", index=False)
```

Conclusion

The experiment successfully analyzed vegetable price data from the AGMARKNET dataset to understand market trends. It compared daily prices with MSP to evaluate farmer profitability. The relationship between arrivals and prices was studied to identify supply-demand effects. A linear regression model was applied to predict future prices, supporting data-driven agricultural decisions.

In [ ]:

```

```

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